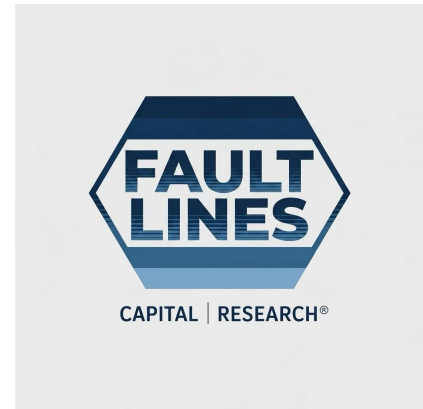




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The Economics of Private Equity Leveraged Roll-Ups in SME acquisition structure:

Efficiency , Risk & Consolidation

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The Economics of Private Equity Roll-Ups: Efficiency , Risk & Consolidation .

ABSTRACT

Over the past two decades, private equity firms have increasingly shifted from traditional *leveraged buyouts* toward highly engineered consolidation strategies known as *roll-ups* — the systematic acquisition and integration of fragmented *small and medium-sized enterprises (SMEs)* within a single vertical. While advocates portray roll-ups as vehicles for operational efficiency, *procurement optimization*, and *EBITDA* expansion, critics argue that many modern roll-up structures are less a triumph of managerial innovation and more an exercise in *multiple arbitrage*, financial engineering, and *market power aggregation*.

At the center of the contemporary roll-up model lies a sophisticated interaction between *platform acquisitions*, *add-on acquisitions*, and *valuation multiple expansion*. A private equity sponsor typically acquires a “*platform company*” at a moderate enterprise value/*EBITDA multiple* and subsequently acquires smaller competitors at materially discounted multiples. Through *post-merger integration*, centralized procurement systems, *SG&A* rationalization, *ERP standardization*, and *working-capital optimization*, the sponsor attempts to re-rate the consolidated entity at the higher platform multiple during exit. This process, often referred to as *multiple arbitrage*, has become one of the most lucrative yet controversial mechanisms in *alternative asset management*.

However, beneath the appearance of operational synergy lies a more complex economic reality. Many roll-ups rely heavily on *leveraged recapitalization structures*, covenant-light debt, and aggressive assumptions regarding *synergy realization*. In fragmented industries such as *HVAC services*, dental clinics, veterinary practices, industrial distributors, *scaffolding manufacturers*, and *valve & fittings supplies*, private equity

informational inefficiencies and succession gaps among founder-led businesses. The resulting consolidation can produce substantial short-term gains in *free cash flow generation* and *enterprise valuation*, but may simultaneously create hidden systemic vulnerabilities, including excessive *leverage accumulation*, integration diseconomies, *supplier concentration risk*, and declining service quality.

INTRODUCTION

When I first began studying private equity roll-ups, I assumed they were merely sophisticated expansion strategies — financially aggressive perhaps, but ultimately productive mechanisms for scaling *fragmented businesses*. The deeper I examined the architecture of modern roll-ups, however, the more I realized that their true power lies not simply in operational efficiency, but in their ability to engineer valuation itself. What appears externally as “growth” is often a carefully constructed system of multiple arbitrage, *leverage optimization*, and *consolidation economics* operating beneath the surface of ordinary industries.

I became particularly interested in how private equity firms quietly transform fragmented sectors — *HVAC contractors*, *industrial distributors*, dental practices, scaffolding manufacturers, and valve suppliers — into highly centralized corporate structures through platform-and-add-on acquisition models. In many cases, these industries are not technologically revolutionary, nor do they attract mainstream attention. Yet they possess the precise characteristics that make them *ideal roll-up targets: recurring cash flows*, low digital penetration, founder-led succession gaps, and *regional fragmentation*. To me, this raises a fascinating question: are modern roll-ups genuinely creating economic productivity, or are they primarily extracting value through *financial engineering* mechanisms that remain poorly understood outside institutional finance?

The answer, I believe, lies in the interaction between operational integration and capital structure manipulation. A *private equity sponsor* may acquire a platform company at a premium *EBITDA multiple*, purchase smaller competitors at *discounted valuations*, centralize procurement systems, *standardize ERP infrastructure*, optimize working capital cycles, and subsequently re-rate the consolidated entity at a materially higher *enterprise value*. On paper, the process resembles rational economic consolidation. In practice, however, I found that many roll-ups depend heavily upon covenant-lite debt structures, aggressive synergy assumptions, and *short-term EBITDA accretion* strategies designed to maximize exit valuation rather than *long-term industrial resilience*.

Through this paper, I aim to examine private equity roll-ups not merely as acquisition strategies, but as instruments of structural economic transformation. By analyzing *multiple arbitrage*, *debt layering*, operational convergence, regulatory blind spots, and

the long-term consequences of consolidation, I seek to explore whether roll-ups represent a *sustainable evolution of capitalism* — or simply one of its most sophisticated forms of financial extraction.

Case Study: The Historical Evolution of Private Equity in SMEs — From Family-Owned Fragmentation to Financialized Consolidation .

I. The Pre-Private Equity SME Economy (1950s–1980s)

For much of the twentieth century, SMEs functioned primarily as independently operated, regionally embedded businesses. In sectors such as *manufacturing, industrial distribution, logistics, construction supply, auto services*, and medical practices, ownership structures were highly decentralized. Most firms were:

- 1) founder-led,
- 2) family-managed,
- 3) underleveraged,
- 4) operationally independent.

Capital allocation decisions were typically conservative. Growth depended upon retained earnings rather than *institutional financing*. Importantly, these businesses were not viewed as scalable financial assets because their value was deeply tied to *localized relationships*, tacit knowledge, and owner reputation.

During this period, private equity itself remained relatively small. *Early leveraged buyouts* focused largely on undervalued public corporations rather than fragmented SMEs. However, two structural conditions were slowly forming that would later make SMEs *highly attractive acquisition targets*:

Industry Fragmentation

Thousands of sectors remained regionally dispersed with no dominant national players.

Succession Vulnerability

Many founder-led firms lacked generational continuity, creating future liquidity opportunities.



Michael Milken : The Junk Bond King

Courtesy: The Gentleman's Journal

II. The Rise of Leveraged Buyouts and Financial Engineering (1980s)

The 1980s marked the institutional birth of modern private equity through the rise of *leveraged buyouts (LBOs)*. Firms such as KKR pioneered large-scale debt-financed acquisitions using *high-yield “junk bond” financing* developed by *Michael Milken*.

The core logic of LBOs was straightforward:

- 1) **acquire companies using substantial leverage,**
- 2) **improve cash flows,**
- 3) **generate amplified equity returns.**

Initially, *SME sectors were not the central focus*. Yet private equity firms increasingly recognized that fragmented middle-market industries possessed several advantages over large public corporations:

- 1) **lower acquisition competition,**
- 2) **limited analyst coverage,**
- 3) **inefficient operations,**
- 4) **and cheaper valuation multiples.**

III. The Roll-Up Boom of the 1990s

The 1990s witnessed the first major wave of *SME-focused roll-ups* across the United States. Private equity sponsors and acquisitive operators targeted *fragmented sectors* such as:

- 1)funeral homes,
- 2)waste management,
- 3)auto dealerships,
- 4)office supply distributors,
- 5)healthcare clinics,
- 6) industrial services.

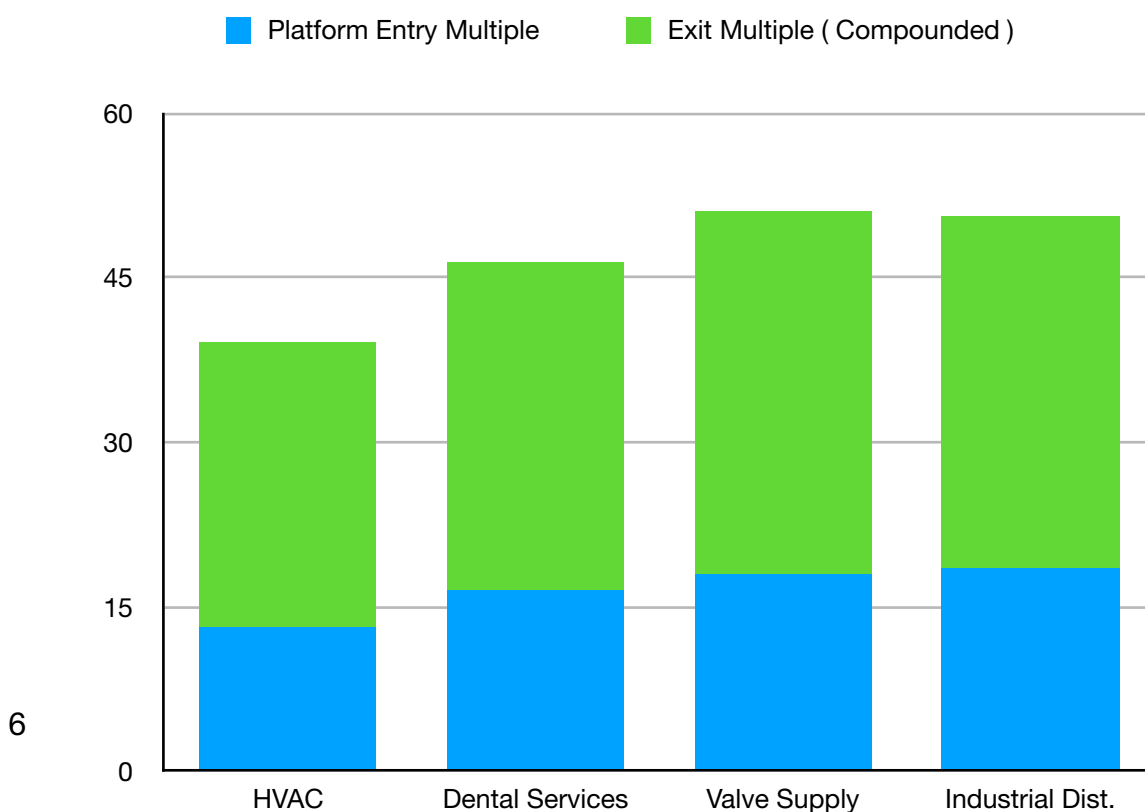
The economic mechanism was revolutionary in its simplicity.

A sponsor would:

- 1)Acquire a “platform company.”
- 2)Purchase smaller local competitors at lower EBITDA multiples.
- 3)Integrate accounting, procurement, branding, and operations.
- 4)Revalue the larger consolidated entity at a higher institutional multiple.

This became known as *multiple arbitrage*.

Fig.1 Bar Graph Illustrating Entry & Exit Multiples of P.E Firms in SME's (1990-2020)



IV. Institutionalization of SME Private Equity (2000s)

By the early 2000s, private equity had evolved from *opportunistic dealmaking* into a *mature institutional asset class* supported by:

- 1) pension funds,
- 2) sovereign wealth funds,
- 3) university endowments,
- 4) insurance capital.

As competition for *large public buyouts* intensified, firms increasingly moved downstream into the middle market.

SMEs became ideal targets because they offered:

- 1) fragmented competition,
- 2) recurring cash flow,
- 3) low technological sophistication,
- 4) operational inefficiencies suitable for standardization.

At the same time, *globalization* intensified pressure on smaller firms. Many independent manufacturers and distributors struggled with such financial jargons.

Fig.2 Lehman Brothers i.e The Global Financial Housing Crisis 2008

Courtesy ; The Wall Street Journal



V. The Modern Era: Hyper-Scaled Roll-Ups (2010's- Present)

The modern roll-up era differs from earlier consolidation waves in one major respect:

technology and data infrastructure now *drive consolidation alongside finance*.

Contemporary private equity firms increasingly use:

- 1)AI-enabled procurement systems,
- 2)predictive maintenance analytics,
- 3)centralized CRM platforms,
- 4)cloud ERP integration,
- 5)algorithmic pricing systems across acquired businesses.

Industries once considered “*boring*” — *dental clinics, veterinary chains, plumbing services, industrial distributors, scaffolding firms, and valve suppliers* — have become *prime institutional acquisition targets*.

This has created what many economists describe as:

- 1)**sub-threshold consolidation**: where acquisitions individually appear insignificant but collectively reshape markets.
- 2)**localized polarisation** : where regional competition gradually disappears without triggering traditional antitrust scrutiny.

Critics argue that roll-ups:

- 1)prioritize short-term EBITDA optimization,
- 2)suppress labor bargaining power,
- 3)increase systemic leverage risk,
- 4)reduce entrepreneurial independence,
- 5) concentrate market power invisibly.

The healthcare sector illustrates these tensions clearly. Private equity acquisitions of dental practices, emergency clinics, and veterinary hospitals have produced:

- 1)standardized operations,
 - 2)centralized billing,
 - 3)aggressive cost optimization,
- but also allegations of declining service quality and *over-financialization*.

Thus, the *SME economy* increasingly sits at the intersection of:

- 1)financial capitalism,
- 2)industrial consolidation,

3) technological centralization.

Conclusion

The history of *private equity in SMEs* is ultimately the story of how *fragmented local businesses* became *institutional financial assets*. What began as *isolated leveraged* buyouts evolved into *sophisticated consolidation ecosystems* driven by multiple arbitrage, *debt structuring*, operational standardization, and *data integration*.

Today, private equity no longer merely invests in SMEs — it actively reorganizes entire sectors through *acquisition architectures* that reshape competition, labor, and ownership itself.

The central question is no longer whether private equity can create efficient businesses. It is whether the long-term economic and social consequences of *financialized consolidation* outweigh the efficiencies it claims to produce.

When Private Equity Went Wrong – The Collapse of Toys “R” Us

One of the most cited examples of private equity failure is the collapse of *Toys “R” Us* after its *leveraged buyout* by a *consortium* including *KKR* and *Bain Capital* in 2005. The acquisition relied heavily on *leveraged financing*, leaving the company burdened with billions in *debt obligations*.



Fig.3 Toys “R” Us : A Lesson for all

Retailers . Courtesy : Babson College

Instead of *directing capital* toward *e-commerce modernization* and operational reinvention during the rise of *online retail*, a substantial portion of *cash flow* was diverted toward *debt servicing* and *interest payments*. The company struggled to compete with *digitally advanced retailers* while simultaneously carrying an unsustainable capital structure.

By 2017, *Toys “R” Us* filed for *bankruptcy*, leading to the closure of hundreds of stores and the loss of tens of thousands of jobs globally. Critics argued that excessive leverage, *dividend extraction*, and *short-term financial engineering* accelerated the company’s decline. The case became symbolic of the darker side of private equity — where *EBITDA optimization* and *leveraged recapitalization* overshadow long-term operational sustainability.

Equation 1 — The Multiple Arbitrage Value Creation Identity

Let V_{exit} be the exit enterprise value, V_{platform} the platform acquisition cost, and the spread between add-on acquisition multiples and exit multiples the core value driver:

$$V_{\text{exit}} = (EBITDA_{\text{platform}} + \sum EBITDA_{\text{add-on}}) \times M_{\text{exit}} \quad \Delta V = V_{\text{exit}} - [V_{\text{platform}} + \sum (EBITDA_{\text{add-on}} \times M_{\text{add-on}})] - \Delta \text{Debt} - \Delta \text{CapEx}$$

Where $M_{\text{exit}} > M_{\text{add-on}}$ is the multiple arbitrage condition. The equity return is entirely a function of this spread, not operational alpha alone.

The company's difficulties intensified after the **2008 financial crisis** as consumer behavior shifted increasingly toward **online purchasing and big-box retail** competition. Although Toys "R" Us continued generating significant revenue, its debt burden severely constrained adaptability. **By 2017, the company filed for bankruptcy protection, eventually leading to widespread liquidation**, store closures, and the loss of tens of thousands of jobs globally.

Toys "R" Us, Inc. and Subsidiaries			
Consolidated Balance Sheets			
Condensed and adapted for educational use			
In millions			
	Fiscal year ending		
	1/28/2017	1/30/2016	
ASSETS			
Current assets:			
Cash and cash equivalents	\$ 566	\$ 680	
Accounts and other receivables	255	225	
Merchandise inventories	2,476	2,270	
Other current assets	92	113	
Total current assets	\$ 3,389	\$ 3,288	
Property and equipment, net	3,067	3,163	
Other assets	452	459	
Total assets	\$ 6,908	\$ 6,910	
LIABILITIES AND EQUITY			
Current liabilities	\$ 2,738	\$ 2,798	
Long-term debt	4,642	4,612	
Other non-current liabilities	820	765	
Total liabilities	\$ 8,200	\$ 8,175	
Total stockholders' equity (deficit)	(1,292)	(1,265)	
Total liabilities and equity	\$ 6,908	\$ 6,910	

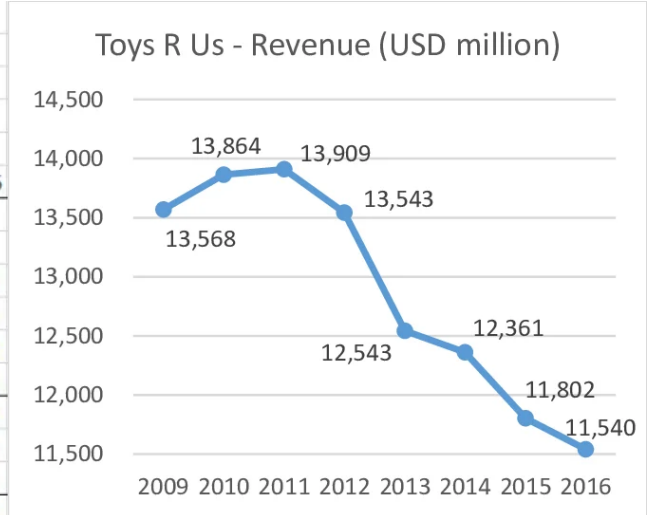


Fig.5 Toys "R" Us Revenue Projection (2009-2016)

Courtesy: ResearchGate

Fig.4 Toys "R" Us Consolidated

Balance Sheets

(2016-2017) .Courtesy:A/C in the Headlines

Equation 2 — The Roll-Up MOIC Decomposition Formula

$$\text{MOIC} = (\text{Revenue_growth} \times \text{Margin_expansion} \times \text{Multiple_re-rating}) / (\text{Leverage_cost} \times \text{Integration_drag})$$

More formally:

$$\text{MOIC} = [(\text{EBITDA}_0 \times (1+g)^n \times \text{M_exit}) / (\text{EBITDA}_0 \times \text{M_entry})] \times (1 / (1 + r_d)^n)$$

Where **g** = blended organic + inorganic EBITDA growth, **r_d** = effective cost of leveraged debt, and **n** = holding period. This decomposition isolates exactly how much of

MOIC is attributable to multiple expansion versus genuine earnings growth — a distinction critics argue PE sponsors deliberately obscure in LP reporting.

Literature Review: Financialization of Fragmented Industries

The rise of private equity roll-ups reflects a broader transformation within modern capitalism: the increasing *financialization* of traditionally operational industries. Financialization refers to the growing dominance of *financial motives*, markets, and institutions in shaping economic activity. While earlier industrial expansion depended largely upon organic growth, contemporary *private equity models* increasingly rely on *acquisition-driven consolidation* financed through *leverage and institutional capital*.

In fragmented industries, private equity firms identify sectors characterized by low technological integration, regional competition, *founder-led ownership*, and inefficient procurement structures. These industries are particularly attractive because they allow sponsors to deploy *platform-and-add-on acquisition architectures* capable of generating both operational efficiencies and valuation expansion.

A core mechanism behind roll-ups is *EBITDA multiple arbitrage*. Smaller firms operating independently often trade at lower valuation multiples due to limited scale, management depth, and *liquidity constraints*. After consolidation, however, the combined entity may receive significantly higher institutional valuations due to perceived scalability, *recurring revenue stability*, and *market concentration advantages*. Consequently, value creation in modern private equity increasingly emerges not merely from operational improvement, but from the *reclassification of fragmented businesses* into institutional-grade assets.

However, critics argue that this process can distort underlying economic productivity. In many cases, enterprise value expansion outpaces genuine operational innovation. Some economists describe this phenomenon as “*financial engineering-led productivity*,” where leverage structures and *valuation optimization* become more important than long-term *industrial development*.

Furthermore, scholars have increasingly highlighted the role of hidden market concentration. Since many *add-on acquisitions* fall beneath traditional antitrust reporting thresholds, roll-ups can gradually produce regional oligopolies without attracting significant *regulatory scrutiny*. This process, often referred to as *sub-threshold consolidation*, has become especially prominent within *healthcare services*, industrial

distribution, veterinary practices, construction supply chains, and infrastructure *maintenance sectors*.

The Mechanics of a Modern Roll-Up Strategy

Modern roll-up strategies are built upon a structured acquisition architecture designed to maximize valuation expansion while minimizing *operational fragmentation*. The process generally begins with the acquisition of a “*platform company*” — an established business possessing stable *cash flows*, experienced management, and scalable operational infrastructure.

Once the platform is acquired, private equity sponsors begin pursuing smaller “add-on acquisitions” within the same industry or adjacent markets. These smaller businesses are often founder-led enterprises lacking sophisticated *ERP systems*, centralized procurement networks, or advanced financial management capabilities.

The economic logic of the roll-up depends upon several interconnected mechanisms:

Multiple Arbitrage

The most significant financial driver behind roll-ups is multiple arbitrage. Smaller firms may be acquired at *4x–6x EBITDA multiples*, while larger consolidated firms may trade at *10x–15x EBITDA* due to *institutional investor demand*. By combining smaller firms into a larger entity, private equity firms effectively create valuation expansion independent of proportional operational growth.

Exit Revaluation

After consolidation and EBITDA expansion, sponsors attempt to exit the investment through:

- 1) secondary buyouts,
- 2) IPOs

- 3) strategic acquisitions,
- 4) dividend recapitalizations.

At exit, the consolidated entity is frequently marketed as a *scalable institutional platform* rather than a collection of fragmented businesses.

While theoretically efficient, this model depends heavily upon successful integration and continuous *capital market liquidity*. Failures in cultural integration, technology implementation, or debt servicing can rapidly *destabilize highly leveraged consolidators*.

SBA Loan Structures and Private Equity Exploitation in SME's : An Elaborative Case Study

One of the most controversial developments in lower middle-market finance is the increasing use of *SBA-backed acquisition structures* by private equity operators, search funds, and highly leveraged acquisition entrepreneurs. The *U.S. Small Business Administration* created the *SBA 7(a) loan program* to support entrepreneurship and provide financing access to small business owners who lacked sufficient *collateral* or credit history. Over time, however, sophisticated *acquisition groups* began using these government-backed structures to finance aggressive consolidation strategies within fragmented industries.

Fig.6 United States Small Business Administration

Courtesy:Wikipedia



Under a typical *SBA acquisition model*, a buyer contributes only a small equity portion — often **10%** of the *total purchase price* — while the majority of the transaction is financed through *SBA-guaranteed* senior debt. Since lenders are partially protected by *federal guarantees*, they are willing to extend leverage levels that would otherwise appear risky in conventional *commercial lending environments*.

Private equity-backed operators recognized that this structure could dramatically amplify *return on invested capital (ROIC)*. In fragmented sectors such as *HVAC services*, plumbing contractors, *industrial maintenance firms*, medical clinics, *logistics operators*, and *manufacturing distributors*, sponsors began acquiring *stable cash-flow businesses* using layered financing structures with minimal direct equity exposure. Many of these businesses possessed highly attractive characteristics:

- 1) recurring revenue,
- 2) low customer churn,
- 3) founder-led ownership,
- 4) predictable EBITDA,
- 5) limited technological sophistication.

The acquisition strategy typically followed a familiar roll-up architecture:

- 1) Acquire a “platform” SME using SBA-backed debt.
- 2) Improve EBITDA through operational optimization.
- 3) Pursue add-on acquisitions using additional leverage.
- 4) Refinance the consolidated entity into conventional commercial debt at a higher valuation multiple.

% Allocation Breakdown : Leveraged SME Acquisition Structure

FINANCIAL COMPONENT	PERCENTAGE ALLOCATION	PURPOSE
SBA-Backed Senior Debt	55%	Primary acquisition financing supported by federal guarantees
Sponsor Equity Contribution	10%	Direct capital invested by the PE sponsor or acquisition entrepreneur
Seller Financing	15%	Deferred payment financed by the business seller

FINANCIAL COMPONENT	PERCENTAGE ALLOCATION	PURPOSE
Credit Facility	5%	Working-capital financing for inventory and operational liquidity
Earn-Out Structure	5%	Future performance-based deferred compensation to seller
Subordinated Debt	10%	Higher-yield secondary leverage used to increase acquisition capacity

Fig.7 SBA (7A) Loan Percentage Allocation Table.

This stacked *leverage model* demonstrates how *modern SME acquisitions* frequently depend more upon debt engineering than *direct ownership capital*. While these structures can generate exceptionally high returns during successful expansions, they simultaneously increase:

- 1) refinancing dependency,
- 2) interest-rate sensitivity,
- 3) liquidity fragility,
- 4) systemic leverage risk.

Consequently, many economists argue that modern *lower middle-market acquisitions* increasingly resemble *scaled-down leveraged buyouts* rather than traditional entrepreneurial business purchases.

Pie Chart: Ownership & Control Structure in a Private Equity SBA Roll-Up

This pie chart represents how a typical private equity-backed holding company consolidates multiple SME subsidiaries acquired using SBA-backed financing structures and leveraged acquisition tactics:

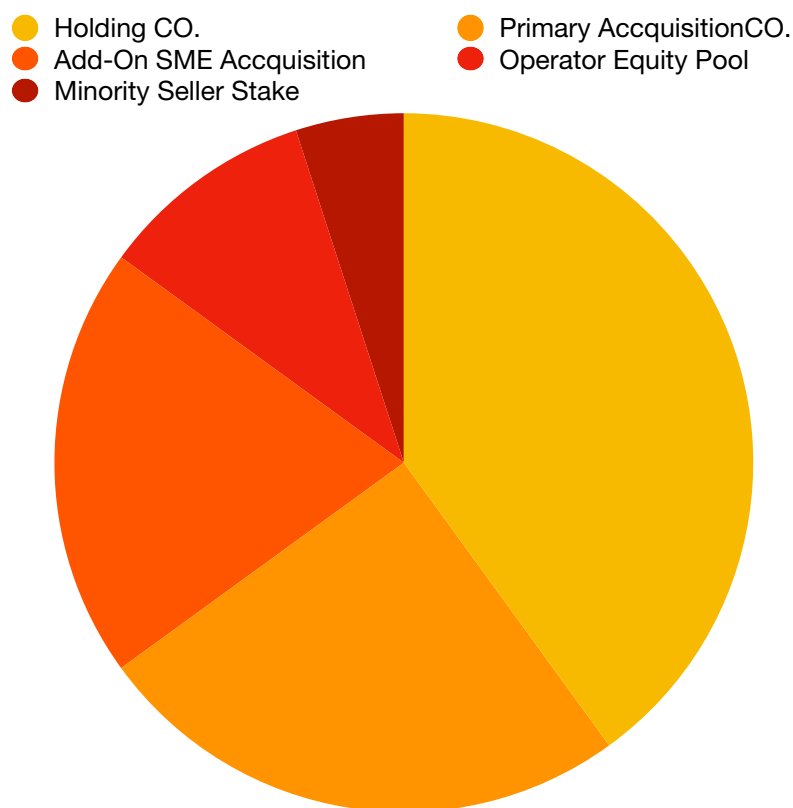


Fig.8 Pie Chart Representing Ownerships in SME Subsidiaries using leveraged SBA-Backed Financing Structures.

Holding Company — 40%

The holding company acts as the *centralized control entity* through which the private equity sponsor governs *capital allocation, debt structuring, operational integration, and acquisition strategy*. This entity often owns *intellectual property, centralized ERP systems*, procurement contracts, and treasury functions.

Platform Company — 25%

The platform company serves as the primary operational foundation of the roll-up. It is usually the first major *acquisition possessing stable EBITDA*, scalable infrastructure, and experienced management.

Add-On SME Acquisitions — 20%

These are smaller *founder-led businesses acquired* after the platform transaction. They are integrated into the broader ecosystem to generate:

- 1)multiple arbitrage,
- 2)procurement efficiencies,
- 3)geographic expansion,
- 4) market concentration

Management / Operator Equity Pool — 10%

Senior *executives and operators* are often granted *equity incentives* to align operational performance with sponsor returns. This structure encourages aggressive *EBITDA growth* and *acquisition integration*.

Minority Seller Retained Stake — 5%

Some founders *retain minority ownership* positions post-acquisition through *rollover equity* agreements, allowing them to participate in *future valuation appreciation* after *consolidation*.

Analytical Insight

This ownership structure demonstrates how modern *private equity roll-ups centralize* operational control through *holding-company architectures* while gradually absorbing *fragmented SMEs* into a single institutional platform. Although individual acquisitions may appear independent, *centralized governance* increasingly concentrates:

- 1)procurement power,
 - 2)pricing influence
 - 3)operational data,
 - 4)regional market control
- within the holding entity itself.

Conclusion

Private equity roll-ups represent one of the *most influential yet controversial mechanisms* of *modern financial capitalism*. What initially emerged as a strategy for consolidating fragmented industries has evolved into a highly sophisticated system of open *leverage optimization, operational integration, and valuation engineering*. Through platform-and-add-on acquisition architectures, private equity firms have transformed thousands of *founder-led SMEs* into *institutional financial assets* governed by *centralized holding-company structures*.

Supporters of private equity argue that roll-ups *improve operational efficiency*, accelerate technological modernization, and solve succession challenges facing aging business owners. In many industries, *fragmented enterprises* lacked the scale necessary to implement *ERP systems*, procurement optimization, predictive analytics, and digitized management infrastructure independently. From this perspective, consolidation can increase productivity and create scalable operational ecosystems.

However, this paper demonstrates that the modern roll-up model extends far beyond simple operational improvement. The widespread use of multiple arbitrage, covenant-lite leverage, *SBA-backed acquisition financing, mezzanine debt layering*, and *EBITDA-focused restructuring* reveals the growing dominance of financial engineering within SME ownership structures. In many cases, enterprise value expansion appears increasingly dependent upon capital structure optimization rather than proportional productive innovation.

The growing use of *SBA-backed loans* within *acquisition-driven consolidation* strategies further exposes the tension between *public credit systems* and *private capital incentives*. Programs originally designed to support entrepreneurial ownership are now, in certain cases, indirectly facilitating *leveraged consolidation* and *hidden market concentration* within fragmented industries.

Furthermore, the rise of *sub-threshold acquisitions* raises significant regulatory concerns. Because many acquisitions occur beneath *conventional antitrust* reporting thresholds, private equity firms can gradually accumulate regional pricing power and operational dominance without *substantial regulatory intervention*. This process challenges traditional definitions of competition and market concentration in the modern economy.

Ultimately, private equity itself is neither inherently destructive nor universally beneficial. Its long-term impact depends upon whether capital is used to build *durable operational* value or merely extract short-term financial returns through *leverage amplification* and aggressive restructuring. The central question is therefore not whether private equity creates value, but rather: who captures that value, how sustainably it is created, and at whose expense the risks are ultimately borne.

“When debt becomes the strategy rather than the tool, fragility becomes inevitable.”
— Ray Dalio

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